

The Better Molecule

Bio-isobutanol for petrol blending and aviation fuel — half of ethanol's mileage penalty, 25% more jet per tonne

Two problems ethanol cannot solve

India's blending programme is hitting a molecule limit, not a capacity limit

The mileage wall

Ethanol's ~34%/L energy deficit is the whole Volume Dividend: E20 already costs drivers ₹22,700 cr/yr in extra litres, and E30 doubles it. Engines are calibrated for E20 — the octane offset is spent. Every further ethanol point is a bigger hidden levy.

The overcapacity trap

~2,000 cr L of distillery capacity chasing ~1,100 cr L of E20 demand; only ~60% of offered ethanol absorbed; utilisation stuck at 65–75% for three years (CareEdge). More ethanol capacity is the last thing India needs.

And a third: aviation has no ethanol answer

SAF mandates arrive from 2027. Ethanol-to-jet wastes carbon: 0.60 t distillate per tonne. The alcohol India has surplus of is the wrong alcohol for the market that is about to be created.

- The same fermentation assets can make a different, better molecule — that is the entire thesis

Isobutanol: a C4 alcohol that behaves like petrol

Same sugar, same fermenters, different organism — and materially better fuel properties

Property	Ethanol	Isobutanol	Why it matters
Energy, MJ/L	21.1	26.5	+26% per litre
Blend dilution at 20%	−6.9%	−3.5%	HALF the mileage penalty
ATJ / SAF yield (t per t)	0.60	0.75	+25% more jet fuel
Water miscible?	yes	no	pipeline-shippable, no phase separation
Blending vapour pressure	raises RVP	neutral / lowers	easier summer-grade compliance

IB20 delivers the same 20% renewable volume at 3.5% dilution instead of 6.9% — the consumer keeps half the energy ethanol takes away

Not a lab curiosity: Gevo and Butamax have run commercial retrofits abroad; the chemistry, the ASTM D7566 ATJ pathway and the engine data all exist. India has none.

The gap that stops everything: no price, no standard

Ethanol has six administered slabs and a BIS blending standard. Isobutanol has neither.

	Ethanol	Isobutanol
Administered OMC price	6 feedstock slabs (₹57.97 – ₹71.86/L)	NONE
BIS blending standard	IS 15464 / IS 2796	NONE
Interest subvention (DFPD ISS)	6%, 1,212 projects sanctioned	NOT eligible
Result	1,970 cr L built	zero litres

The ask to policy

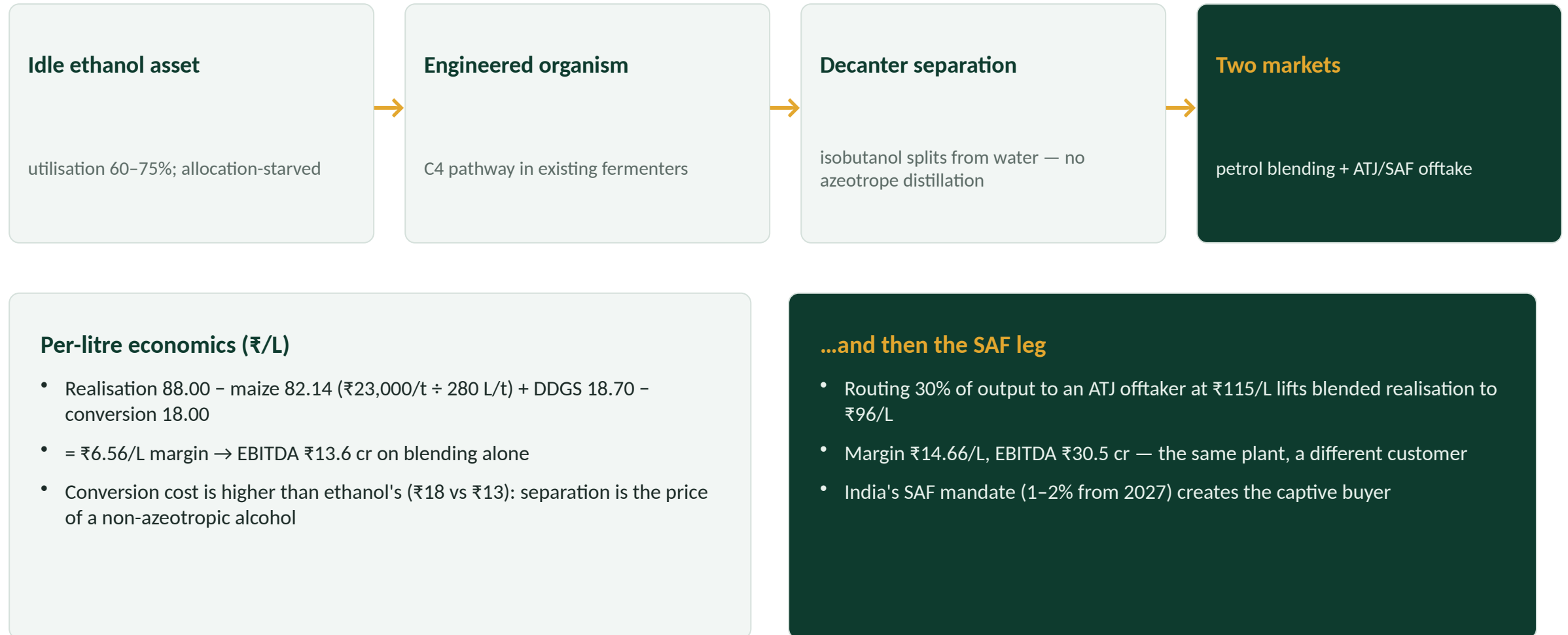
- Price isobutanol on ENERGY, not litres: parity with the maize ethanol slab = ₹90.3/L. We ask ₹88 — below parity, so the OMC buys renewable energy cheaper than it does today

Why per-litre pricing kills it

- Isobutanol yields ~280 L/t of maize vs ethanol's ~400. Paid at ethanol's per-LITRE slab the margin is -₹9.58/L — a guaranteed loss. Energy-basis pricing is not a subsidy; it is the correct unit.

Business model: retrofit idle ethanol capacity

100-KLPD grain distillery → 70 KLPD isobutanol • ₹80 cr retrofit • the overcapacity IS the opportunity



Financials: blending alone is marginal — SAF makes it a business

₹80 cr retrofit · 70:30 @ 9.5% · NO subvention assumed (isobutanol is not ISS-eligible)

Output routed to ATJ/SAF	Blended realisation ₹/L	Margin ₹/L	EBITDA ₹ cr	Project IRR
0% — blending only	88.0	6.56	13.6	13.8%
30% — base case	96.1	14.66	30.5	33.9%
50% — SAF-led	101.5	20.06	41.7	45.6%

Sensitivity — project IRR (%): maize price × isobutanol realisation (blending only, no SAF)

Maize ₹/t	₹80/L	₹88/L (ask)	₹96/L
20,000	21.1	39.6	56.4
23,000 (base)	loss	13.8	33.6
26,000	loss	loss	5.0

Read the 'loss' cells honestly: without energy-basis pricing AND cheap grain, blending-only isobutanol does not work. The SAF leg is what carries it.

Risks and mitigants

This is a technology-and-policy bet — we say so plainly

Risk	Reality	Mitigant
Technology (the big one)	No Indian bio-isobutanol plant operates; yield 280 L/t and ₹18/L conversion are the least certain numbers here	Licence a proven organism (Gevo/Butamax-class); pilot train before full retrofit; capex staged on demonstrated yield
No price slab	Sold at ethanol's per-litre slab the margin is – ₹9.58/L	Energy-basis pricing is the pre-condition, not a hope: no notification, no capex
No BIS standard	IS 15464 covers ethanol only	Standards ask filed alongside the price ask; ASTM D7566 + global engine data support it
SAF timing	The 1-2% mandate lands 2027; ATJ offtakes are nascent	Blending-only case still positive (13.8%) at base — SAF is upside, not survival
Feedstock	Same maize exposure as any grain distillery	Multi-feed front end retained; DDGS credit larger per litre than ethanol's

- Not claiming: ISS eligibility, carbon credits, or that this is as bankable as a molasses-to-grain conversion (it is not — it is the higher-return, higher-risk sibling)

Landscape: proven abroad, absent in India

The window is the gap between India's SAF mandate and its ethanol glut

Technology holders

Gevo (US) and Butamax (BP/DuPont JV) have operated commercial isobutanol retrofits of corn-ethanol plants — the exact play proposed here. Licensing, not invention, is the route in.

Indian adjacency

Praj has an ATJ/SAF programme and builds the distilleries; IOCL and Indian Oil–LanzaJet activity signals OMC appetite for alcohol-to-jet; the 2027 SAF mandate is the demand anchor.

The opening

India has 900+ cr L of idle distillery capacity, a maize-slab price that already tolerates ₹71.86/L alcohol, and no isobutanol at all. First mover sets the standard and the slab.

- Fits the platform logic of the other pitches: CBG monetises waste carbon, multi-feed conversion monetises seasonality — isobutanol monetises the ethanol glut by turning surplus fermentation into a higher-value molecule

The ask: one licensed pilot, gated on policy

₹80 cr

retrofit (₹24 cr equity)

23 kt/yr

isobutanol from idle capacity

33.9%

project IRR at 30% SAF routing

2 asks

energy-basis slab + BIS standard

- Gate 1 (policy): an energy-basis OMC price notification for isobutanol (₹88/L against the ₹90.3 parity) and a BIS blending specification — no notification, no capex. This is the pitch's honest precondition, and the reason nothing exists today
- Gate 2 (technology): licence a proven organism and demonstrate ≥ 280 L/t on Indian maize in a pilot train before the full ₹80 cr retrofit is drawn
- Then: one ATJ/SAF offtake for 30% of output; replicate across the idle grain-distillery fleet — the same 900+ cr L of stranded capacity the ethanol overcapacity analysis identifies
- Why back the harder molecule: it is the only route that raises blending WITHOUT raising the consumer's hidden levy, and the only alcohol that gives aviation 25% more fuel per tonne of Indian carbon